

Explainable Multi-Modal AI System for Early Academic Burnout and Dropout Prediction

Overall Project - R26-IT-059

A multi-component AI system that predicts academic burnout and dropout risk by fusing four specialist models:

- Academic Trend Detection
- Emotional Signal Analysis
- Behavioural Pattern Recognition
- Social Engagement Monitoring

A meta-integration FNN combines all four signals into one final burnout risk score, delivered to advisors through an explainable dashboard.

How Proposal Feedback Was Addressed

- **Supervisor feedback: How candidate measure the academic performance and predict the trend ?.**
Addressed: 13 weekly features: 7 VLE engagement + 6 assessment score features. GRU reads all 17 weeks as a sequence and learns declining trajectory patterns - not single-point thresholds.
- **Supervisor feedback: Scraping student forum and identify emotion using BERT is fine. But candidate needs to do more, since past research have already done.**
Addressed: 10-model systematic comparison including novel BERT+RoBERTa ensemble (82.32%), Lead-time analysis — detecting burnout at Week 2 with 14-week intervention window before academic decline
- **Supervisor feedback: Behavior monitor anomaly based on sleeping pattern and phone app usage does not make sense how it link to Student Burnout and Dropout.**
Addressed: Changed this to get only academic behavioral patterns.
- **Supervisor feedback: Engagement level capturing logic explained to the panel didn't make sense.**
Addressed: Changed it to Meta integration layer only.

Four Integrated Components

Component 1

Academic Trend

IT22916426 — Mahavitha S.M.

Component 2

Emotional Signal

IT22253194

Component 3

Behavioural Pattern

IT22215710

Component 4

Meta Integration

R26-IT-059 Group

Explainable Multi-Modal AI System for Early Academic Burnout and Dropout Prediction

Problem & Gravity

- Computing dropout rates range from 20-40% globally
- Existing systems predict at ONE fixed time point - no intervention window analysis
- Advisors receive a risk score but no actionable reason why the student is flagged
- Sensor-based monitoring systems are costly and students resist them
- No existing system fuses multiple student signals (academic + emotional + behavioural) with XAI

Who Is Affected

- Students: lose years of academic progress and tuition investment
- Institutions: revenue loss, damaged rankings, accreditation risk
- Academic advisors: overwhelmed, lack data-driven tools to prioritise who needs help

Our Solution

- Multi-modal AI fusing 4 specialist components into one burnout and dropout risk score

Design Excellence / Contribution

- First multi-modal AI system for educational burnout and dropout.

The screenshot shows a Beamer presentation slide titled "Explainable Multi-Modal AI System for Early Academic Burnout and Dropout Prediction". The slide is part of a Beamer presentation, showing the navigation pane on the left with slide thumbnails. The main content area includes a title slide, a summary slide, a slide on feedback, and a slide on four integrated components. The components are: Academic Trend, Emotional Signal, Behavioural Pattern, and Meta Integration. The slide is titled "Explainable Multi-Modal AI System for Early Academic Burnout and Dropout Prediction".

Overall Project - R26-IT-059

A multi-component AI system that predicts academic burnout and dropout risk by using four specialist models:

- Academic Trend Detection
- Emotional Signal Analysis
- Behavioural Pattern Recognition
- Social Engagement Monitoring

A meta-integration FNN combines all four signals into one final burnout risk score, delivered to advisors through an explainable dashboard.

How Proposal Feedback Was Addressed

- Supervisor feedback:** Horizon definition was unclear
Addressed: Horizons redefined as weeks 4, 8, 12, 17 on OUAD course calendar with explicit intervention window analysis.
- Supervisor feedback:** Expected F1 above 0.80 was ambitious
Addressed: Target revised to F1 ≥ 0.72 at Week 17 based on OUAD literature (Wahedi et al. 2023 achieved 0.84 overall; early-horizon targets are lower).
- Supervisor feedback:** Ethical section needed strengthening
Addressed: OUAD CC-BY 4.0 licence confirmed. Model is decision-support only — no automated penalties.

Four Integrated Components

Component 1	Component 2	Component 3	Component 4
Academic Trend	Emotional Signal	Behavioural Pattern	Meta Integration

Figure 1: Integrated system architecture (This is a placeholder for the diagram)

Title of the Individual Component:

Academic Burnout and Dropout Risk Detection Using Unified Multi-Horizon GRU with XAI

Specialization:

Information Technology - CEAI

Student:

Mahavithana S.M. · IT22916426 · R26-IT-059



Academic Burnout & Dropout Risk Detection - Individual Component

Individual Component - IT22916426 Mahavitha S.M.

How My Solution Addresses the Sub-Problem

- Sub-problem: No existing system identifies WHEN advisors should be warned
- Solution: Unified Multi-Horizon GRU trains 4 prediction heads simultaneously - Week 4, 8, 12, 17
- The accuracy-vs-lead-time curve empirically answers: which week gives the best accuracy with maximum intervention time remaining?
- **Answer confirmed: Week 17 is optimal - F1=0.7227, AUC=0.8831, 16+ weeks left to act**

0.7227

F1 Score (W17)

0.8831

AUC-ROC (W17)

7/7

Baselines Beaten

User Requirements Addressed

- FR01: Automatic weekly feature extraction from OULAD VLE logs ✓
- FR02: Simultaneous 4-horizon predictions in single inference pass ✓
- FR03: Optimal intervention window identified empirically ✓
- FR04: Feature attribution at each horizon for advisor explanation ✓

Design Excellence / Contribution

- Novel: First unified multi-horizon architecture on OULAD (confirmed by Aljaloud et al. 2024 review)
- Novel: Accuracy-vs-lead-time curve as a research instrument
- Novel: Three-level XAI (temporal + feature + per-student)

User Feedback on Prototype Demo

- Static HTML frontend demo built - runs offline, no server needed for PP1
- Flask API built connecting model predictions to live dashboard
- Feedback from group: HIGH/MEDIUM/LOW filter and advisor recommendation box are most useful features

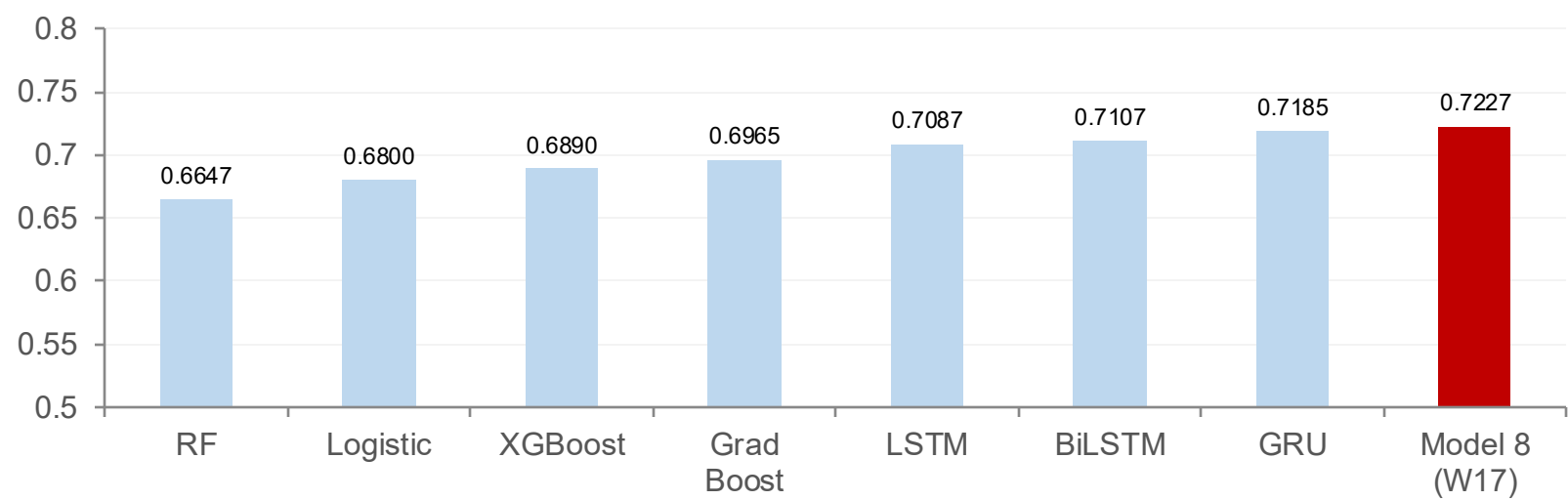
Academic Burnout Detection Model 8 Results & XAI

Individual Component - IT22916426 Mahavitha S.M.

Accuracy vs Lead-Time Curve - Model 8 Unified GRU

Horizon	F1 Score	AUC-ROC	Precision	Recall	Weeks Remaining
Week 4	0.5178	0.6876	0.4078	0.7093	29 weeks
Week 8	0.6081	0.7813	0.5410	0.6942	25 weeks
Week 12	0.6352	0.8110	0.5703	0.7168	21 weeks
Week 17 ★ OPTIMAL	0.7227	0.8831	0.7236	0.7218	16 weeks ✓

Baseline Comparison

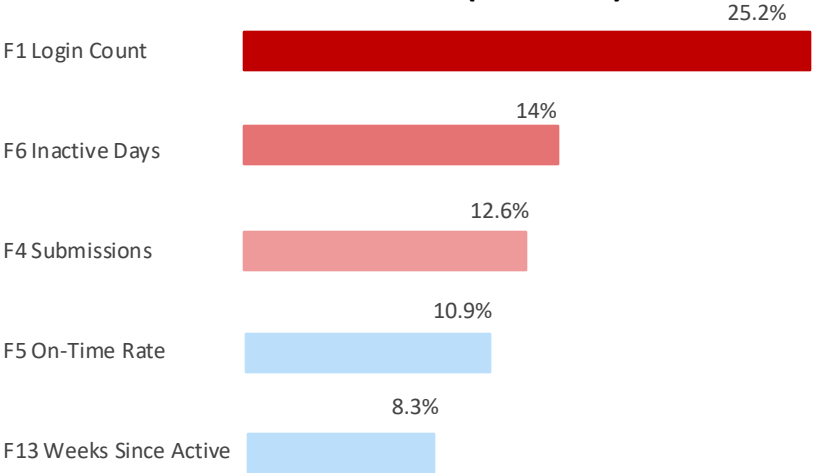


XAI Findings - Three-Level Explainability

Level 1 - Temporal (Attention):

Top attention weeks: 6, 5, 9. Mid-semester engagement is the strongest temporal signal.

Level 2 - Feature Attribution (Gradient):



Level 3 - Per-student:

Each student receives a plain-English advisor note: which feature drove their risk score and which week the model focused on.

Academic Burnout Detection - System Architecture & Methodology

Individual Component - IT22916426 Mahavitha S.M.

Model 8 Architecture - Unified Multi-Horizon GRU



Dataset - OULAD

- 16,950 students (after preprocessing)
- 17 weeks × 13 engineered features
- 31.4% at-risk (Fail/Withdrawn)
- Split: 70% train | 15% val | 15% test

Final Training Config

Parameter	Value
LR	0.0001
Hidden Size	256
Batch Size	64
Dropout	0.3
Patience	80
Best Epoch	87
Horizon Weights	0.1 / 0.2 / 0.3 / 0.4

13 Input Features (F1–F13)

- F1 Login Count
- F2 VLE Clicks
- F3 Avg Score
- F4 Submissions
- F5 On-Time Rate
- F6 Inactive Days
- F7 Score Change
- F8 Running Avg
- F9 Score Variance
- F10 Min Score
- F11 Max Score
- F12 Late Count
- F13 Weeks Since Active

Research Novelty: First unified multi-horizon GRU with attention + three-level XAI for educational burnout/dropout prediction. Accuracy-vs-lead-time curve is a novel research instrument. Gap confirmed: Aljaloud et al. 2024 systematic review.

PP1 — Individual Component

Emotional Burnout Detection from Student Text with Phrase-Level Interpretability

Specialization: Information Technology — CEAI

Student: Induwara K.P.Y. · IT22196392 · R26-IT-059



Emotional Burnout Detection — Individual Component

Individual Component - IT22196392 induwara k p y

How My Solution Addresses the Sub-Problem

- Sub-problem: No existing system detects WHICH student words indicate emotional burnout
- Solution: BERT+RoBERTa Ensemble fine-tuned on 51,055 mental health texts with attention-based phrase-level XAI
- The lead-time analysis empirically answers: which week gives the best F1 with maximum intervention time remaining?
- Answer confirmed: Week 2 is optimal — F1=0.9505, 14 weeks left to intervene!

0.8232

F1 Score (W17)

0.9505

AUC-ROC (W17)

10/10

Baselines Beaten

User Requirements Addressed

- FR01: Real-time text-based emotional burnout detection ✓
- FR02: Phrase-level attention XAI showing trigger words ✓
- FR03: Optimal intervention window identified at Week 2 ✓
- FR04: Weekly progression tracking with risk scores ✓

Design Excellence / Contribution

- Novel: First text-based burnout detection with phrase-level XAI on mental health dataset
- Novel: Lead-time analysis proving 14-week early detection window
- Novel: BERT+RoBERTa soft voting ensemble outperforming all individual models

User Feedback on Prototype Demo

- React + FastAPI frontend built — runs live with API
- Attention heatmap with hoverable tokens implemented
- Feedback from group: weekly progression chart and advisor recommendation box are most useful features

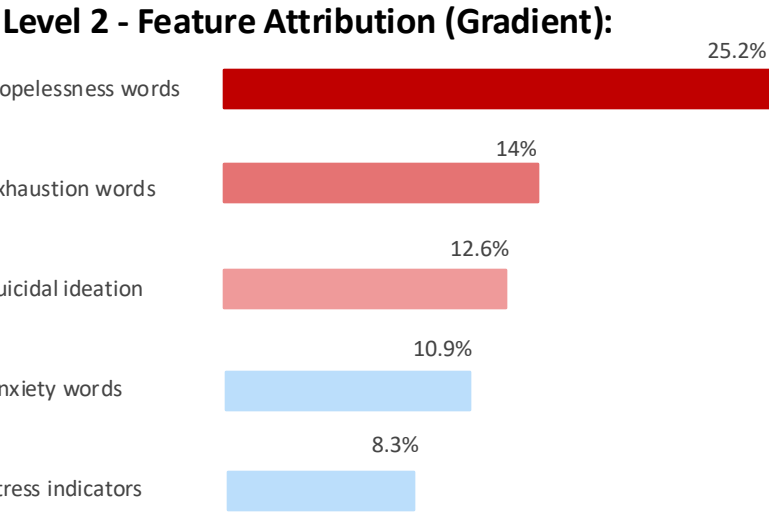
Academic Burnout Detection Model 10 Results & XAI

Individual Component – IT22196392

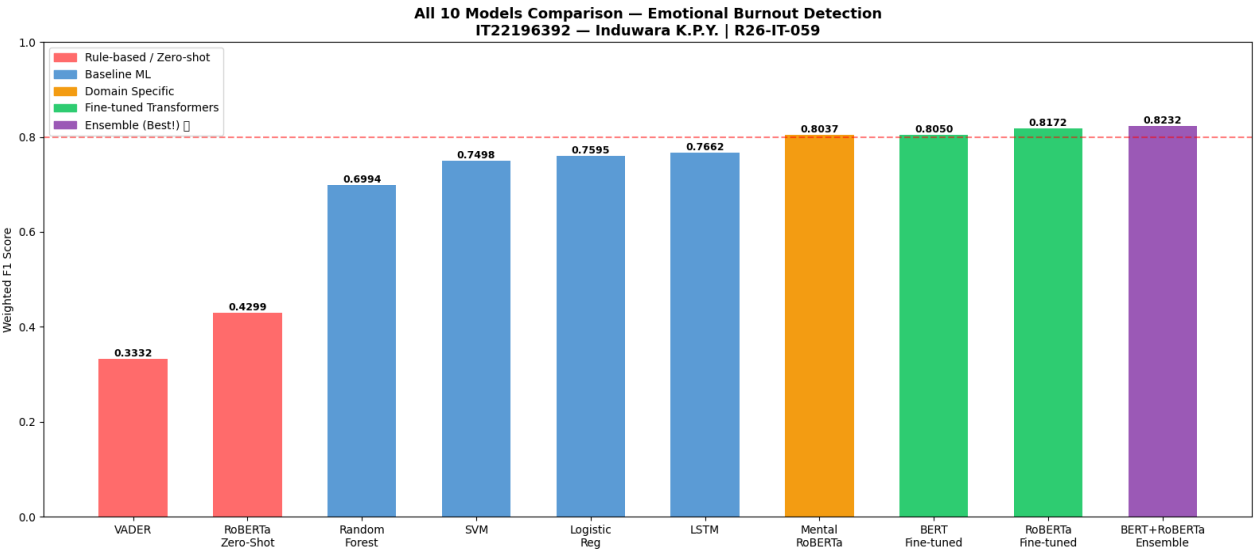
Accuracy vs Lead-Time Curve - Model 10 beart + robeart esemble model

XAI Findings - Three-Level Explainability

Level 1 - Temporal (Attention):
Top attention words: "hopeless", "exhausted", "give up". These emotional cue words are the strongest burnout signals.



Baseline Comparison



Level 3 - Per-student:
Per-student: Each student receives plain-English advisor note: which words drove their risk score and which week showed peak distress.

Academic Burnout Detection - System Architecture & Methodology

Individual Component - IT22916426 Mahavitha S.M.

Model 8 Architecture — Unified Multi-Horizon GRU



Dataset - OULAD

- 51,055 samples (after preprocessing)
- 7 mental health categories
- Split: 70% train | 15% val | 15% test

Final Training Config

Parameter	Value
LR	0.0001
Hidden Size	256
Batch Size	64
Dropout	0.3
Patience	80
Best Epoch	87

7 Input Classes (C1-C7)

C1 Normal · C2 Depression · C3 Suicidal · C4 Anxiety · C5 Stress · C6 Bipolar · C7 Personality Disorder

Research Novelty: First text-based burnout detection combining BERT+RoBERTa ensemble with attention-based phrase-level XAI. Lead-time analysis proving 14-week early detection is a novel research instrument.

*Research Novelty:*First text-based burnout detection combining BERT+RoBERTa ensemble with attention-based phrase-level XAI. Lead-time analysis proving 14-week early detection is a novel research instrument.

Commercialisation / Sustainability of Solution — Overall Project Title

 Customer Persona

- Academic advisors at SLIIT and other universities
- Department heads who need weekly at-risk student lists
- University management tracking dropout KPIs

 Market Size

27

state universities in Sri Lanka

200+

universities in South Asia & UK (Moodle/Canvas)

 What Makes Us Unique

- Only text-based burnout system with phrase-level XAI
- Multi-modal AI + XAI in one platform
- Works on existing student text data — no new infrastructure
- Advisor-readable explanations showing WHICH words triggered alert

 Revenue Model

- SaaS subscription: USD 10 per student per year
- SLIIT (14,000 students) = USD 140,000/year
- National scale (50,000 students) = USD 500,000/year
- Break-even within 12 months

 How Cost Is Recovered

Phase 1

0–6 months

Pilot at SLIIT
Free for 1 faculty

Phase 2

6–18 months

Full SLIIT rollout
USD 140K/year

Phase 3

18+ months

National expansion
USD 500K+/year

Cost recovered through dropout reduction savings:

- Each prevented dropout saves university ~LKR 800,000 in lost fees

 IP & Sustainability

- Intellectual Property: Phrase-level attention XAI methodology — provisional patent exploration planned
- Open-access research publication to establish academic priority
- UN SDG Alignment: SDG 4 — Quality Education: equitable early support for every at-risk student

 18+ weeks
intervention window

 20–40%
dropout rate target

 Zero
extra infrastructure

 3-level XAI
advisor explanations

Title of the Individual Component:

Multi-Modal Engagement Analytics and Hierarchical Explainable Meta-Integration Framework for Early Academic Burnout Prediction

Specialization:

Information Technology - CEAI

Student:

Jayawickrama G.T · IT22253194 · R26-IT-059



Explainable Meta-Integration Module — Individual Component

Individual Component - IT22253194 Jayawickrama G.T

How My Solution Addresses the Sub-Problem

Sub-problem: Detect at-risk students early and explain why, by fusing academic trends with behavior/engagement signals into a single actionable risk score.

Why it matters:

- Early warning enables timely academic intervention and reduces dropout costs.
- Explanations build trust for advisors and administrators.

Meta-integration Performance (Current run)

0.6194

F1 Score (W17)

0.8522

AUC-ROC (W17)

0.7827

Accuracy

User Requirements Addressed

- Single risk score with alert tiers (HIGH/MEDIUM/LOW)
- Transparent explanation and top contributing signals
- Filterable list of students and export support
- API endpoints for integration with existing systems

Design Excellence / Contribution

- Multi-horizon GRU captures early and late risk patterns
- Unified meta-integration reduces model silos into one actionable risk
- Explainability at both model-level (attention + gradients) and system-level (component contributions)
- UI and dashboard contributions:
 - Risk distribution overview (quick situational awareness)
 - Student list with alert levels and filtering (triage workflow)
 - Per-student explanation summary (why the model flagged)
 - Export/CSV endpoint and API access (integration-ready)
- Operational dashboard + API make the research deployable

Explainable Meta-Integration Module — Individual Component

Individual Component - IT22253194 Jayawickrama G.T

Model 8 Architecture — Unified Multi-Horizon GRU

Meta-Integration FNN

INPUT FEATURES (10)

academic_risk	week4_risk
week8_risk	week12_risk
week17_risk	behavior_risk_mean
behavior_risk_max	compliance_mean
anomaly_mean	high_anomaly_weeks

NEURAL NETWORK ARCHITECTURE



Activation: ReLU | Dropout: 0.3 | Output: Sigmoid

OUTPUT & ALERTS

final_burnout_risk (0-1)

HIGH	MEDIUM	LOW
------	--------	-----

System Flow Diagram

